

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

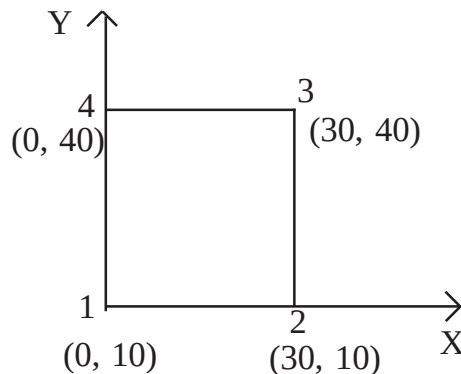
Roll No

ME/IP/PR-701(B) (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Computer Aided Engineering and FEM
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain the terms.
 - a) Constant Strain Triangle (CST)
 - b) Linear Strain Triangle (LST)
 - c) Quadratic Strain Triangle (QST)
2. For the elements shown in figure. Assemble Jacobian matrix and strain displacement matrix for the Gaussian point (0.57735, 0.57735).



3.
 - a) Explain the term Shape function.
 - b) Why polynomial terms are preferred for shape functions in finite element method?
4. Explain the following briefly:
 - a) Effect of element aspect ratio on accuracy
 - b) Numbering nodes for bandwidth minimization
5. Write short notes on followings:
 - a) Hamilton's principle
 - b) Consistent and lumped mass matrices
6. Explain the different commercial software packages used for static analysis of frames.

OR

Discuss in details the solution of simultaneous equations using any one method.

[2]

7.
 - a) Discuss global and local coordinate system.
 - b) What is Banded and skyline assembly?
8. Write a short Note on any three from the following:-
 - a) Axis-symmetric elements.
 - b) Modeling of infinite d.o.f. system into finite d.o.f. system.
 - c) Area co-ordinate system
 - d) Shrink fit simulations
 - e) Lumped mass matrices.
